Gabriel Greenberg

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SUMMARY

Data scientist with 6 years of experience turning environmental sensor data into validated datasets, statistical models, and peer-reviewed research on ozone and other air pollutants. Builds end-to-end Python and R workflows for data pipelines, QC, time-series modeling, and geospatial analysis. M.S. in Data Science (4.0 GPA) and B.A.Sc. in Chemistry; co-author of 5 peer-reviewed publications.

EXPERIENCE

Data Scientist — Boulder Atmospheric Integrated Research (Boulder A.I.R.), Boulder, CO 2020–Present

- Write analysis scripts and run exploratory data analysis on complex, nonlinear atmospheric systems, critically interpreting the results for peer-reviewed papers, city and government reports, and conference presentations.
- Led the analysis for a grant-funded ozone source study (Colorado Air Quality Enterprise) combining atmospheric transport models with 11 years of measurements at 23 sites; first-author interactive poster, Oct 2026; journal manuscript in preparation.
- Built Skytap, a Python pipeline on Linux and Docker that ran 285,103 HYSPLIT back trajectories, temporarily downloading and processing 30 TB of meteorological data in parallel across multiple machines to meet time, memory, and cost limitations.
- Built pscf_py, a multi-site Python library for potential source contribution function (PSCF) analysis, a geospatial statistical method for mapping likely pollution source regions; no multi-site PSCF library existed in Python.
- Rebuilt legacy station-specific submission scripts into a generalized Python library that formats data into EBAS-compliant NASA Ames files for international archives.
- Created a QC and data-finalization toolkit: automated flagging, an interactive plot-based review tool, and before/after checks for Picarro, Thermo NOx, ozone, and VOC instruments.
- Located a natural gas pipeline leak in a North Broomfield neighborhood (June 2022) by tracing ethane spikes to a plume with an ethane/methane ratio 4–5× that of other directions and no acetylene signal; the City of Broomfield used these leads to find and shut in the leak.
- Built and maintain the company website, bouldair.com, and maintain IDAT, the company's live Plotly dashboard.

Research Intern — Institute of Arctic and Alpine Research (INSTAAR), University of Colorado Boulder Summer 2019

- Ran GC-FID analyses of global flask samples to measure non-methane VOCs and wrote Python scripts to plot and interpret their trends.

PROJECTS

Isoprene Forecasting — M.S. Capstone, UW–Eau Claire · abstract 2025

- Forecast isoprene 24 hours ahead from Boulder A.I.R. measurements merged with drought, vegetation, and weather data; removed daily and seasonal cycles with FFT and Butterworth filters.
- Compared SARIMA, XGBoost random forest, and PyTorch neural network models tuned with Bayesian optimization; the neural network cut test error about 50% versus SARIMA and came close to a published physics-based model.

Allylic — chemistry education site · Next.js, React Three Fiber · allylic.org

- Designed and built a public chemistry learning site with interactive 3D molecule views.

EDUCATION

M.S., Data Science — University of Wisconsin–Eau Claire · GPA 4.0 May 2025

B.A.Sc., Chemistry concentration — Quest University Canada, Squamish, BC · President's Achievement List April 2020

PUBLICATIONS

Co-author of 5 peer-reviewed papers (2024–2026) in *J. Geophys. Res. Atmos.*, *Elementa*, *ACS ES&T Air*, and *J. Air Waste Manag. Assoc.* on ozone, VOCs, methane, and airborne radioactivity near oil and gas development. Full list

SKILLS

- **Languages:** Python, R, SQL, Bash, JavaScript/TypeScript, HTML/CSS
- **Python:** pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost, PyTorch, Matplotlib, Plotly, Dash
- **Modeling:** time series (SARIMA, FFT filtering), regression, random forests, neural networks, Bayesian optimization, geospatial statistics (PSCF)
- **Data engineering:** Linux, Docker, Git, SQLite/MySQL, Spark, SSH automation
- **Atmospheric:** HYSPLIT, GC-FID, ozone, NOx, methane, VOCs, PM2.5